

BANK OF AMERICA · ENGINEERING LEADERSHIP

The Angular 14 → 18 upgrade, run as a pipeline instead of a project

Cognition · Devin

WHAT WE ALREADY KNOW ABOUT YOUR ENGINEERING

~\$14B

annual technology budget — about \$4B of it on new initiatives

60,000

people in the technology organisation

~59M

verified digital users · ~30B digital interactions last year, up 14%

86%

of clients rate their digital experience 9 out of 10 or better

18,500

developers on GitHub Copilot — up from 500 in under a year

You are not early on AI in the SDLC. You have already done the hard part: enterprise-wide developer adoption.

American Banker, Jun 2026 (budget, headcount, Copilot 500 → 18,500) · Banking Dive, Jun 2026 · BofA newsroom, Mar 2026.

SO WHAT IS ACTUALLY MISSING

“We have 17,000 programmers using AI coding technology today **saving 10% to 15% in code generation costs.**”

Brian Moynihan, Q2 2025 earnings call

Code generation is the cheap part of an Angular major upgrade. The expensive parts are deciding whether a diff is safe, knowing why an override existed, and telling twenty consuming teams their build is about to break.

An assistant makes a developer faster inside a file. This work is measured in completed, reviewable pull requests across repositories.

WHY THIS USE CASE, OF YOUR THREE

Angular upgrade

Risk is **fan-out**. One shared library, a custom design system over Material, consumed by teams who did not ask for this upgrade. Failure arrives as somebody else's broken build in deadline week.

Cloud migration

Risk is runtime — and you already own the harder half: a private cloud you burst out of by design, not a lift-and-shift.

Test coverage

Risk is measurable, and the examination hands you the number to hit.

Fan-out is the risk no dashboard shows you — and the only one where an agent working across repositories changes the shape of the problem.

Angular 14 left LTS in June 2024. Angular 18 reached end of life in November 2025.

The stated target no longer satisfies the policy that created the deadline.
Supported today: 20 (to Nov 2026), 21, 22.

**If one hop is a project, you will land on an unsupported version again.
The deliverable isn't 18 — it's a hop you can run four times a year.**

THE DEMO — A REAL ANGULAR 14.2 NX MONOREPO

13 Material components wrapped as bofa-*, 18 documented overrides, three consuming applications, SSO/MFA and analytics stubbed at their boundaries. Live at t1mchee.github.io/devin-pitch

- The customer surface, so the pixels mean something.
- Two gates: one tells you **something moved**, one tells you **which promise broke**.
- Two independent Devin runs of the same hop — and what both of them got wrong.

An override with no recorded reason *is* the migration risk

```
/* OV-05c: header cells carry the slate-900 brand weight,  
   not Material's grey — brand review, Mar 2023 */  
.bofa-table .mat-header-cell { color: $boa-slate-900; }
```

Material renames the internal class between majors. The selector dies silently. The intent — a brand decision someone signed off — is now unenforced, and nothing in the repo knows.

Migrating a design system is migrating intent, not selectors.

TWO GATES

Pixels

21 snapshots, budget 40 absolute pixels.

```
753 pixels differ on table-default
```

Intent

77 computed-style probes — one per override reason — plus a legibility sweep scoring contrast on every visible text node, 16 routes, both palettes.

```
0V-05c: header cells carry the  
slate-900  
brand weight, not the Material grey
```

| One is a diff. The other is a sentence a reviewer can act on.

WHERE IT FAILED — THE SLIDE THAT SHOULD BUY YOUR TRUST

- The pixel budget was 0.1% of pixels. It waved a **wrong brand red through to a customer dashboard**. Now 40 absolute pixels.
- The first dark-theme check asserted the light-theme colour: two of five transaction rows rendered at **1.07:1** — unreadable balances — and the suite was green.
- The intent gate's first run found **three dead overrides in our own library**; one was passing only because Material's default happened to equal the brand colour.
- A migration run explained its own red screenshots as a missing-font problem. It wasn't. **The gate held; the narrative failed.**

Found by adversarial review, not by design — and all of it written down in the repository.

WHAT THE AGENT DID, AND DID NOT, DO

- Two independent runs of the same playbook both left CI **red** on the snapshots Material's MDC rewrite changed — and **neither regenerated a baseline** to get green.
- Both stopped on the same ambiguous override with the same reasoning. That reasoning was **wrong**; human review caught it, not a test.
- Against a **third-party** design system with zero intent comments: 3 genuine stops, 1 false stop, and three confident wrong guesses no gate caught.

It does not guess when it knows it doesn't know. It does not yet know when a confident rewrite has quietly stopped applying — so the intent gate is pilot scope, not a nice-to-have.

CONTROLS, IN YOUR LANGUAGE

- The output is a **pull request**. It inherits every control you already have — CODEOWNERS, branch protection, your scanners.
- **No baseline change without a written reason.** CI rejects the PR body without BASELINE-CHANGE:, and the guard cannot be edited by the PR it guards.
- `npm ci --ignore-scripts` as policy: no transitive lifecycle script runs before a human reads the diff.
- Renderer pinned by digest, so “it passed on my machine” is not evidence.
- Every session is a replayable transcript: prompt, commands, diffs, approvals.

“Who changed the customer-facing control, on what evidence, reviewed by whom?”

Done by hand, that question is answered by a commit message. Here it is answered by a transcript, a named override intent, a measured contrast ratio and a review record — for every hop, every time.

The audit artefact is a by-product of the work, not a document written afterwards.

Two weeks · one repository · one named design-system owner

- **Week 1** — land the intent gate on the real library, run 14 → 15 on a branch, open three consumer PRs.
- **Week 2** — run the next hop, measure reviewer minutes per PR, publish the refusal rate.
- **Success** — n consumer PRs merged with no baseline regenerated without a written reason.
- **Failure we would accept** — it stops more often than it ships, and the transcripts tell you exactly why.

THE ASK

- Name the repository and the design-system owner.
- One 90-minute working session with the consuming teams' leads.
- Read-only passes on the other two initiatives in parallel, at no risk.

You have already proven your developers will adopt this. The open question is whether the work arrives finished and reviewable. Two weeks and one repository answers it.